

Sujay Shankar

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🌐 github.com/Sujay-Shankar

Education

The University of Texas at Austin

Bachelor of Science: Astronomy

Bachelor of Science: Computational Physics

Certificate: Elements of Computing

August 2020 – December 2023

Major GPA: 3.8927

Major GPA: 3.9058

Certificate GPA: 4.0000

Boston University

Doctor of Philosophy: Astronomy (in progress)

September 2024 – present

Research Projects

The University of Texas at Austin | McDonald Observatory

May 2022 – August 2023

Undergraduate Research Assistant

Austin, TX

- Lead developer of the `gollum` Python library, analyzing and visualizing stellar and substellar atmosphere models
- Software architecture improvements, UI/UX improvements, and bug fixes
- Added support for starspot two-component mixture modeling with PHOENIX
- Tested functionality on IGRINS spectra

The University of Texas at Austin | Department of Astronomy

August 2023 – December 2023

AST 375C: Conference Course in Astronomy

Austin, TX

- Lead developer of `blase3D`, a fork of `blase` (Gully-Santiago & Morley 2022)
- Used interpretable machine learning with GPUs to clone PHOENIX spectra across T_{eff} , $\log(g)$, and $[\text{Fe}/\text{H}]$
- Used linear interpolators to create manifolds mapping stellar properties to line-by-line properties

The University of Florida | Department of Astronomy

May 2023 – present

REU Student Researcher

Gainesville, FL

- Synthesized a globular cluster escapee sample from APOGEE DR17 and GALAH DR3, combined with Gaia dynamics
- Developed a multithreaded orbit integration pipeline with Monte Carlo initial conditions
- Used chemical, dynamical, and photometric information to match escapee candidates with globular clusters

The University of Texas at Austin | Department of Astronomy

January 2024 – July 2024

Research Engineering/Scientist Assistant

Austin, TX

- Added `gollum` support for newly released Sonora Diamondback brown dwarf atmospheric models
- Improved `gollum`'s documentation, setup, testing, and directory management systems
- Submitted papers for `blase3D` to ApJ and `gollum` to JOSS

Publications

- Shankar, S. & Gully-Santiago, M. & Morley, C. 2025 (in review) *A New Hybrid Machine Learning Method for Stellar Parameter Inference*. ApJ
- Shankar, S. & Bandyopadhyay, A. & Ezzeddine, R. (in prep) *Tagging Globular Cluster Escapee Candidates from APOGEE and GALAH*. ApJ
- Shankar, S. et al. 2024 *gollum: An intuitive programmatic and visual interface for precomputed synthetic spectral model grids*. JOSS

Technical Skills

Languages: Python, Bash, MATLAB, L^AT_EX, Swift, Julia, Lua

Frameworks: Pandas, Altair, PyTorch, SWMF(BATSRUS, FLEKS)

Contexts: Data Analysis/Visualization, High-Performance Computing, Software Packaging, Multi-Platform Development

Conferences

American Astronomical Society 243rd Meeting

January 2024

– Poster: *Novel Dynamical Tagging of Globular Cluster Escapee Candidates back to their Sources*

New Orleans, LA

2023 Bash Symposium

October 2023

– Poster: *Precision Fundamental Stellar Properties with Interpretable Machine Learning*

Austin, TX

TACCSTER 2023

October 2023

– Attendee Only

Austin, TX

Presentations

Dynamically Tagging Globular Cluster Escapee Candidates back to their Sources

August 2023

Generating Rotational Velocities for 27 Near-IR Objects

May 2023